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**ENBC455 – Bioinformatics Engineering**

**05/08/2026**

# Quantify A Transcript Project

1. Experimental information

a. Project title

Common pathogenic mechanisms in the hippocampus across neurodegenerative dementias: Alzheimer's disease, Down syndrome, and Parkinson's disease (human)

b. BioProject accession number

PRJNA1419230; GEO: GSE318560

c. All related SRX accession numbers

Alzheimer's:

- SRR37123805
- SRR37123806
- SRR37123807
- SRR37123808

Non-Demented (Control):

- SRR37123815
- SRR37123816
- SRR37123817
- SRR37123818

d. Sequencing platform

GPL24676 Illumina NovaSeq 6000 (Homo sapiens)

e. Project overview (in your own words)

The following research project “Common pathogenic mechanisms in the hippocampus across neurodegenerative dementias: Alzheimer's disease, Down syndrome, and Parkinson's disease (human)”, examines whether the three neurodegenerative dementias, Alzheimer's disease, Parkinson's disease dementia, and Down syndrome dementia share a common underlying mechanism in the hippocampus. By performing RNA-sequencing on the human brain tissue, they found across all three conditions, they shared 45 differentially expressed genes. The researchers ended up highlighting EHMT2 and LMNB2, which were significantly downregulated in all the dementia groups compared to healthy controls. The study suggested that the two genes that they found can be universal biomarkers that can be used for research studies.

f. Experimental groups and number of replicates per group

3 Experimental Groups: Down syndrome dementia, Alzheimer’s, Parkinson's disease dementia

1 Control Group: Non-Dementia

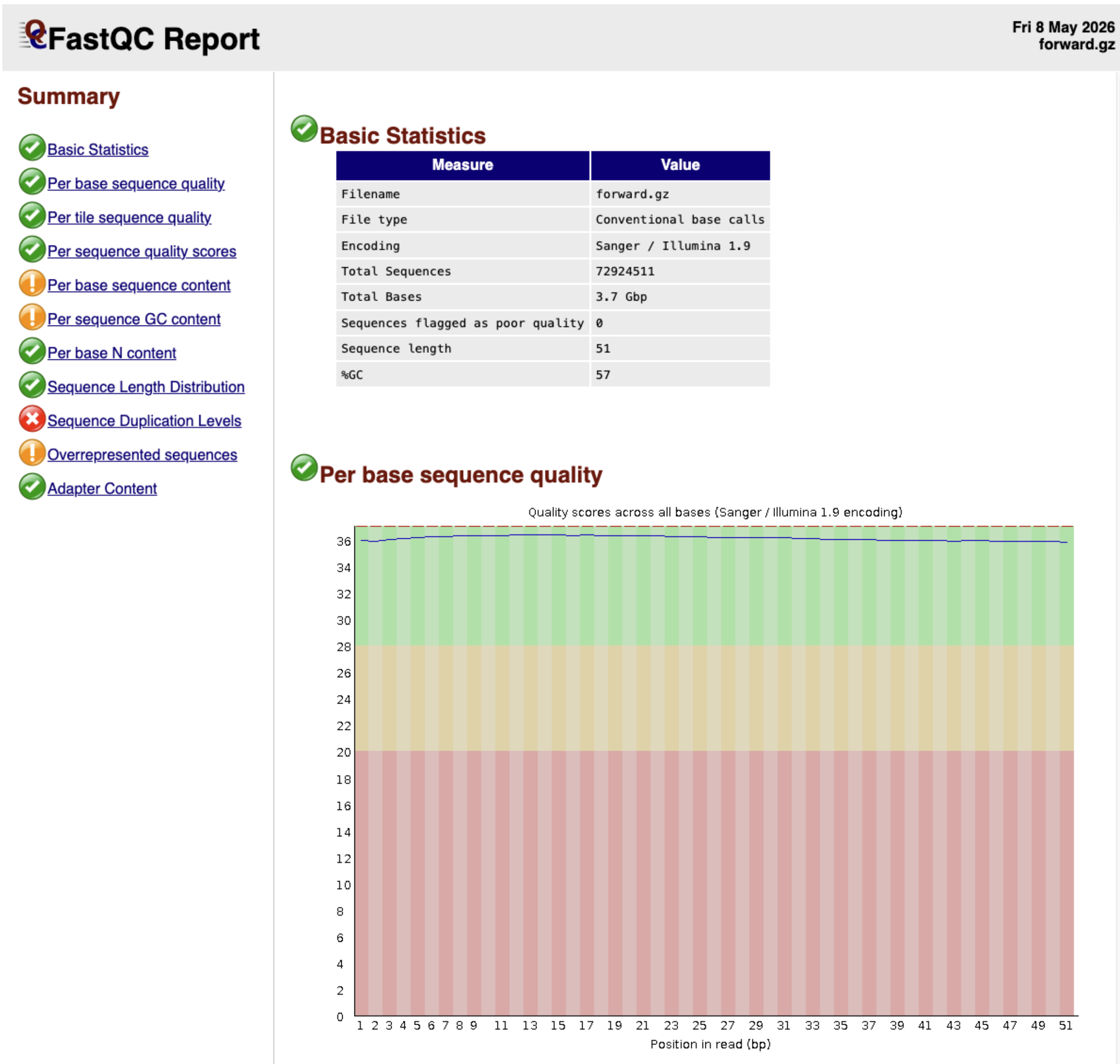
Replicates: 5 Down syndrome dementia, 6 Alzheimer, 4 Parkison's disease, 5 Non-Dementia

g. Hypothesis

Alzheimer’s disease is characterized by severe cellular stress and protein misfolding. As a result, I hypothesize that the hippocampus of Alzheimer's patients will show a significant downregulation in genes specifically associated with protein translation and ribosomal assembly compared to healthy controls.

2. BONUS: you may include the results of any optional processing/quality control steps that you perform, with a summary of why you ran the tool, what you found, and whether you acted on any of the information.

Representative Screenshots of Forward and Reverse Strands



FastQC Report

Fri 8 May 2026
forward.gz

**Summary**

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

**Basic Statistics**

| Measure | Value |
|---|---|
| Filename | forward.gz |
| File type | Conventional base calls |
| Encoding | Sanger / Illumina 1.9 |
| Total Sequences | 72924511 |
| Total Bases | 3.7 Gbp |
| Sequences flagged as poor quality | 0 |
| Sequence length | 51 |
| %GC | 57 |

**Per base sequence quality**

## FastQC Report

Fri 8 May 2026
reverse.gz

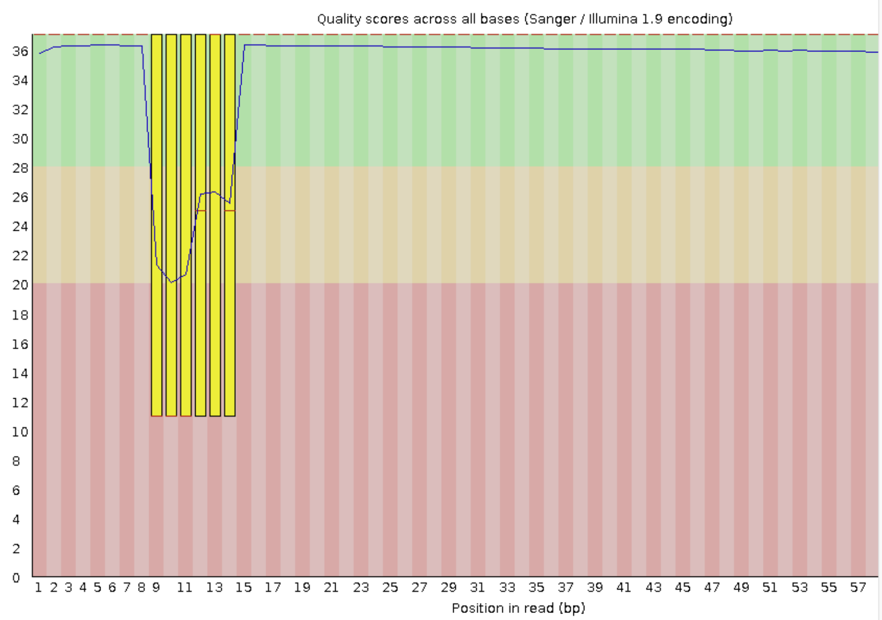
### Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

### Basic Statistics

| Measure | Value |
|---|---|
| Filename | reverse.gz |
| File type | Conventional base calls |
| Encoding | Sanger / Illumina 1.9 |
| Total Sequences | 72924511 |
| Total Bases | 4.7 Gbp |
| Sequences flagged as poor quality | 0 |
| Sequence length | 65 |
| %GC | 55 |

### Per base sequence quality




The provided screenshots of the representative forward and reverse strands show how both are overall great quality. Specifically for the reverse strands, the data quality does dip somewhere around the middle, but this can be due to mechanical errors within the machine itself. This is a probable cause due to all of the other reverse strand positions are very high in quality. In terms of the other statistics, FASTQC was designed specifically to analyze a random even distribution of a whole genome. Obviously, RNA strands were given to test the quality and since RNA has natural biological variations, the potential errors that appear might be due to normal biology not the actual quality of the reads.

3. Mapping, quantification, and normalization

a. Program(s) used for mapping, quantification, and normalization

Salmon quant was used for mapping and the transcript quantification. Normalization was done though DESeq2.

b. Any non-default options selected in these programs

Used GENECODE_Human_Transcriptome (as fasta) imported from the following link, https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44/gencode.v44.transcripts.fa.gz, as the salmon index. Under the Data Input / Is this library mate-paired? Section, the Paried-end Dataset Collection option was chosen.

c. List of the output files generated and which were input into the DE analysis tool

The salmon quant mapping tool generated a single dataset collection consisting of both 4 Alzheimer and 4 control individual tabular output files. The files represented the transcript counts for SRR37123805, SRR37123806, SRR37123807, SRR37123808, SRR37123815, SRR37123816, SRR37123817, and SRR37123818. All eight of these tabular files were subsequently used as the direct inputs for the differential expression analysis tool, DESeq2, where they were grouped into Factor 1 (Alzheimer's disease) and Factor 2 (Control).

4. DE analysis

a. Program used

DESeq2

b. Any non-default options selected in this program

For DESeq2, “Choice of Input data” to Salmon and GENCODE_Human_GTF file imported from the following link, https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44/gencode.v44.annotation.gtf.gz, served as the annotation file. The optional option to output the DESeq2 plots and normalized count tables were selected as well.

c. Results table file (can be downloaded from Galaxy and submitted directly without formatting)

Please see attached

d. Brief research on the differentially expressed RNA features/genes (up to 10)

i. What genes/ncRNAs are represented? Were the up- or downregulated? What do these genes/ncRNAs do? Are they related to the study hypothesis?

- ENSG00000184984 (CHRM5): Downregulated (log2FC = -2.70). A protein encoding gene that produces Chrolinergic Receptor Muscarinic 5 which is a receptor involved in synaptic transmission in the nervous system.
- ENSG00000119471 (HSDL2): Downregulated (log2FC = -1.67). A protein encoding gene that possesses oxidoreductase activity and functions within the mitochondria and peroxisomes.
- ENSG00000263465 (SRSF8): Downregulated (log2FC = -1.41). Produces the Serine and arginine rich splicing factor 8, a protein that acts as a pre-mRNA splicing factor to help process RNA transcripts.
- ENSG00000183148 (ANKRD20A2P) & ENSG00000196922 (ZNF252P): The first is highly upregulated (log2FC = 8.13) and the second is highly downregulated (log2FC = -7.92). Both are pseudogenes.

Yes, these differentially expressed genes strongly support the study hypothesis that Alzheimer's disease features widespread structural, regulatory, and translation-based collapse in the hippocampus. The significant downregulation of CHRM5 aligns with the known pathology of Alzheimer's disease, which is famously characterized by the degradation of cholinergic neurons and a loss of acetylcholine signaling. Additionally, the downregulation of HSDL2 points toward mitochondrial dysfunction and oxidative stress, which are well documented drivers of neurodegeneration.

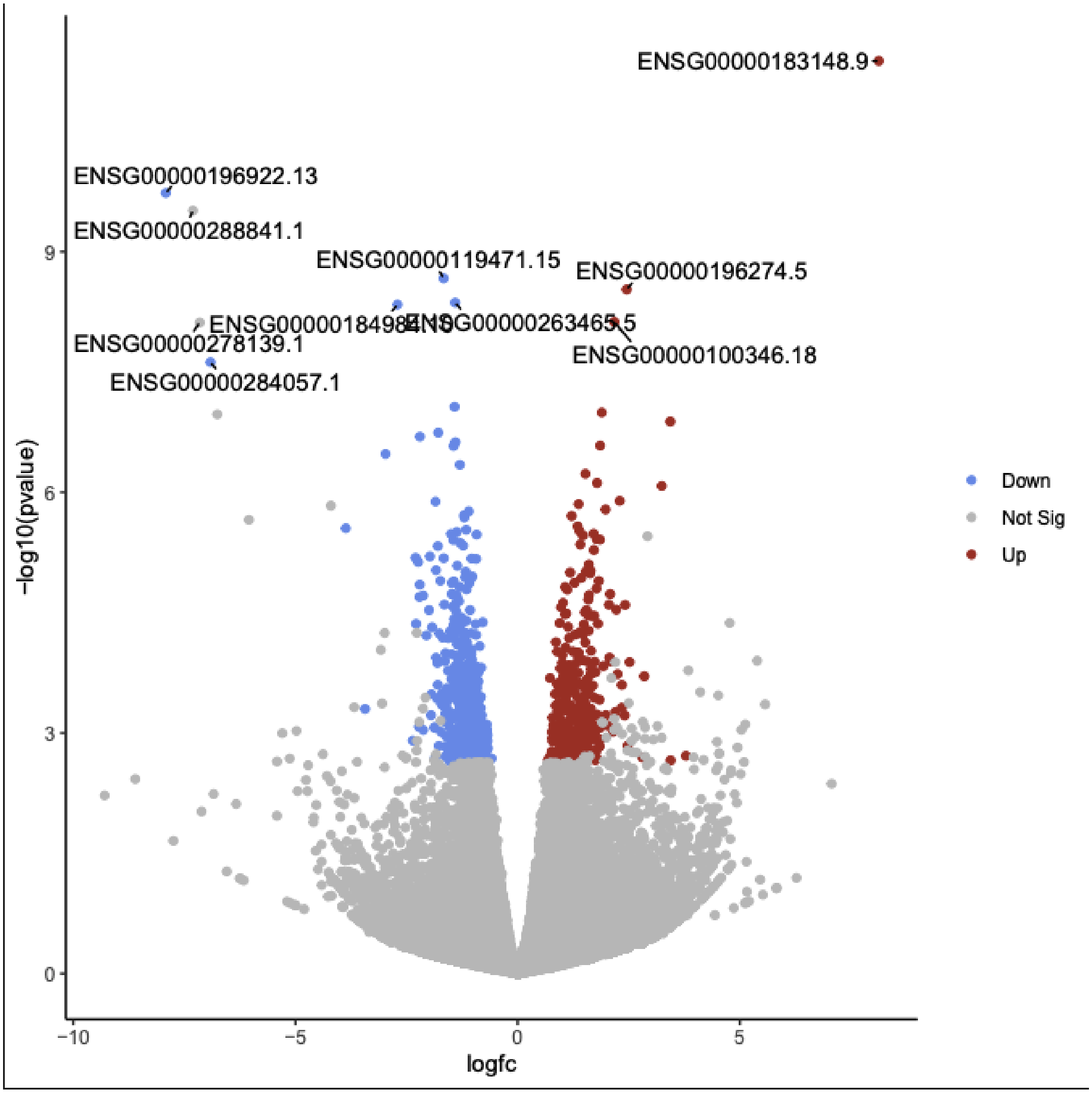
e. Brief summary of each auto-generated plot

i. What does this plot show? What conclusions can you draw from the plot?

- PCA Plot maps the overall variance or differences in gene expression among all samples between Alzheimer's and control. The Alzheimer samples and control samples seperate quite well along PC1 and PC2, which proves that the biological condition of having Alzheimer's is the main driver of the differences in gene expression.
- Sample to sample distances heatmap shows how mathematically similar the overall RNA profiles of the samples are to one another where dark blue means they are highly similar. Samples usually cluster based on their condition but the clustering for this heatmap is not perfect. The bottom right cluster contains three Alzheimer's samples (Datasets 9/10, 11/12, and 15/16, corresponding to SRR37123805, 06, and 08) while the top left cluster contains all four healthy control samples along with one Alzheimer's sample in the middle of them (Datasets 13/14, corresponding to SRR37123807). This would indicate that SRR37123807 is an outlier and where its gene expression might look more like a non-demented brain than an Alzheimer's brain.
- Dispersion estimates plots the variance in gene expression against the mean expression level with the red line being fit to the data. Since the the majority of the black dots are near the line, this means the DESeq2 statistical model was able to successfully fit to the data and controlled the background noise.

- Histogram of p values shows the frequency of different p values across all the genes tested. The massive peak on the left indicates that DESeq2 found a huge number of biologically significant, differentially expressed genes.
- MA−plot for Disease_Status: Alzheimer vs Control plots the magnitude of change against the average expression of the gene with the blue dots being the genes that are statistically significant. Even though most genes are around 0, a clear spread of significant blue dots are scattered above and below the zero line which indicates the up and down regulation of many genes within the Alzheimer's samples.

f. Volcano plot with top differentially regulated RNA features/genes labeled




g. BONUS: optional results from biological interpretation tools

Goseq was used to output a table

i. Auto-generated plots

| Column 1 | Column 2 | Column 3 | Column 4 | Column 5 | Column 6 | Column 7 | Column 8 | Column 9 |
|---|---|---|---|---|---|---|---|---|
| category | over_represented_pvalue | under_represented_pvalue | numDEInCat | numInCat | term | ontology | p_adjust_over_represented | p_adjust_under_represented |
| GO:0005840 | 1.79745715068863e-08 | 0.999999996191352 | 24 | 220 | ribosome | CC | 0.000173317298851024 | 1 |
| GO:0015935 | 2.08944639286435e-08 | 0.999999997591003 | 14 | 77 | small ribosomal subunit | CC | 0.000173317298851024 | 1 |
| GO:0022627 | 2.26479613447631e-08 | 0.999999998260781 | 11 | 45 | cytosolic small ribosomal subunit | CC | 0.000173317298851024 | 1 |
| GO:0002181 | 4.29208190558433e-08 | 0.99999999102272 | 22 | 160 | cytoplasmic translation | BP | 0.000246344040971013 | 1 |
| GO:0003735 | 7.51148569202167e-08 | 0.999999985752339 | 19 | 165 | structural constituent of ribosome | MF | 0.000344897377034867 | 1 |
| GO:0044391 | 1.35825638292376e-07 | 0.999999971708994 | 20 | 186 | ribosomal subunit | CC | 0.000519714167319397 | 1 |
| GO:0022626 | 1.77331903866025e-07 | 0.999999972835156 | 15 | 103 | cytosolic ribosome | CC | 0.000581597978422316 | 1 |
| GO:0032991 | 4.27995479365683e-07 | 1 | 266 | 6088 | protein-containing complex | CC | 0.00122824002690967 | 1 |
| GO:1990904 | 1.86501912822705e-05 | 0.999991052553035 | 50 | 1135 | ribonucleoprotein complex | CC | 0.0475745657175963 | 1 |
| GO:0016282 | 3.33643345139182e-05 | 0.999997869960196 | 6 | 17 | eukaryotic 43S preinitiation complex | CC | 0.0765978391770535 | 1 |

ii. What biological pathways are differentially regulated? Are they related to the experimental hypothesis?

Gene Ontology (GO) over-representation analysis using the goseq tool revealed pathways related to the ribosome (GO:0005840), cytoplasmic translation (GO:0002181), and the small ribosomal subunit (GO:0015935) were significantly over represented among the differentially expressed genes. The results are definitely related to the experimental hypothesis. Alzheimer's disease is biologically characterized by the accumulation of toxic, misfolded proteins. Finding severe dysregulation in the cellular machinery responsible for building proteins (the ribosomes) directly supports the hypothesis that the hippocampus is undergoing translation based stress and structural collapse.

5. Summary

a. Did the workflow work as expected? If not, how would you troubleshoot given more time?

The overall workflow work was very successful, taking the raw sequence data through quantification (Salmon quant) and differential expression analysis (DESeq2). Two areas that did require a lot of troubleshooting was during the optional Gene Ontology analysis. The goseq tool initially failed because the GENCODE gene IDs contained decimals which didn't match the internal database. This was troubleshooter by using the "Replace Text" tool to strip the decimals before running the analysis. If there was more time, something that could be troubleshooted would be the random Alzheimer's sample (SRR37123807) clustered completely with the healthy control group since this did not make any sense biologically.

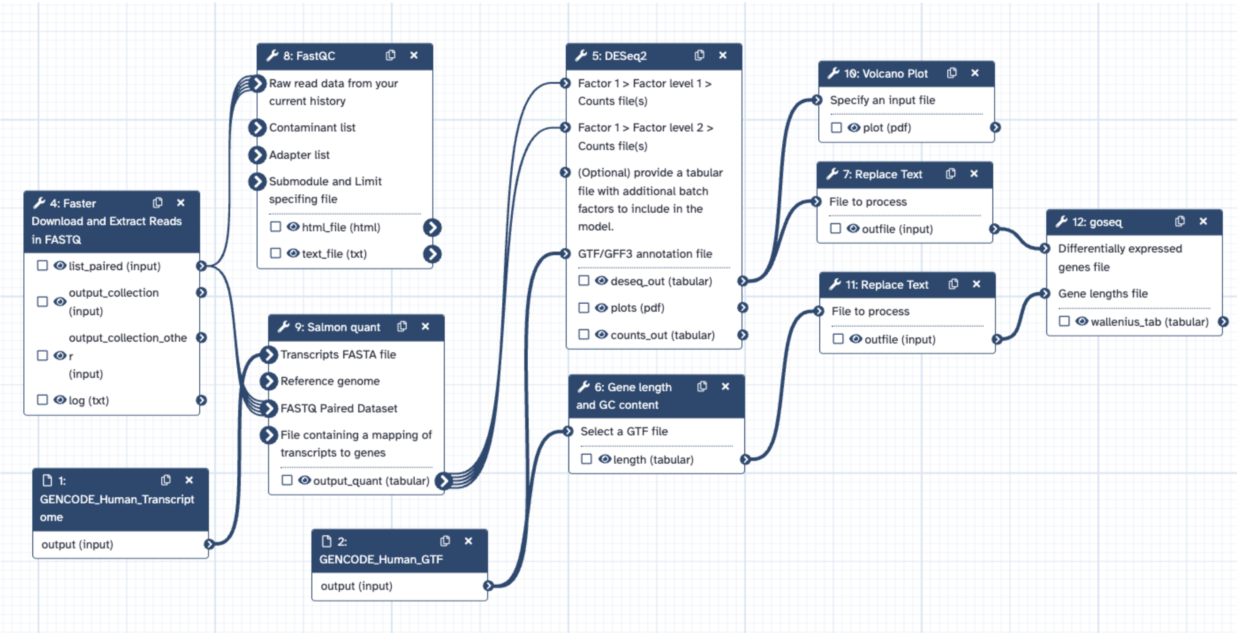
b. What was the outcome of the DE analysis? Is it possible to tentatively accept or reject the hypothesis?

The differential expression analysis successfully identified hundreds of genes with a statistically significant adjusted p-value ($< 0.05$). Notably, key functional genes such as CHRM5 (cholinergic signaling) and HSDL2 (mitochondrial function) were significantly downregulated, while pseudogenes and specific regulatory RNAs were upregulated. Since a clear, statistically significant difference in the gene expressions of the Alzheimer's hippocampus was identified compared to the healthy controls, we can confidently reject the null hypothesis. Furthermore, the biological context of these altered genes (heavy dysregulation in ribosomal and protein translation pathways) allows us to tentatively accept the alternative hypothesis that Alzheimer's pathology is driven by specific, measurable transcriptional changes related to cellular stress and structural collapse.

c. What are the next steps?

Next steps would be to re-run the DESeq2 analysis excluding the outlier sample to ensure the differential expression list is clean and as accurate as possible. Another option could be to use the last sample remaining for Alzheimer's instead and evaluate its effect on the current pipeline. A RT-qPCR on physical samples of the hippocampal tissue can be done to manually confirm the RNA transcript levels of the top highly significant genes.

6. Workflow image showing all programs used and linking the program inputs and outputs appropriately.



8: FastQC
Raw read data from your current history
Contaminant list
Adapter list
Submodule and Limit specifing file
html_file (html)
text_file (txt)
5: DESeq2
Factor 1 > Factor level 1 > Counts file(s)
Factor 1 > Factor level 2 > Counts file(s)
(Optional) provide a tabular file with additional batch factors to include in the model.
GTF/GFF3 annotation file
deseq_out (tabular)
plots (pdf)
counts_out (tabular)
10: Volcano Plot
Specify an input file
plot (pdf)
7: Replace Text
File to process
outfile (input)
12: goseq
Differentially expressed genes file
Gene lengths file
wallenius_tab (tabular)
11: Replace Text
File to process
outfile (input)
4: Faster Download and Extract Reads in FASTQ
list_paired (input)
output_collection (input)
output_collection_other (input)
log (txt)
9: Salmon quant
Transcripts FASTA file
Reference genome
FASTQ Paired Dataset
File containing a mapping of transcripts to genes
output_quant (tabular)
6: Gene length and GC content
Select a GTF file
length (tabular)
1: GENCODE_Human_Transcriptome
output (input)
2: GENCODE_Human_GTF
output (input)

**References**

Crans, R. A. J., Fructuoso, M., Bascón-Cardozo, K., Recaioglu, H., Sotelo-Fonseca, J., Vermeiren, Y., Strydom, A., Dam, D. V., De Deyn, P. P., Rodríguez-Martín, B., Potier, M.-C., & Dierssen, M. (2026, February). *Common pathogenic mechanisms in the hippocampus across neurodegenerative dementias: Alzheimer's disease, Down syndrome, and Parkinson's disease (human)*. National Center for Biotechnology Information. https://www.ncbi.nlm.nih.gov/bioproject/1419230

EBI. (n.d.-a). https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44/gencode.v44.basic.annotation.gtf.gz

EBI. (n.d.-a). https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44/gencode.v44.transcripts.fa.gz

Galaxy. (n.d.). https://usegalaxy.org/